

Apurva Kokate

Machine Learning Researcher & Engineer | PhD Candidate

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Areas of Expertise

Graph Neural Networks • Machine Learning Systems • Scientific & Applied Machine Learning • Deep Learning for Graphical, Vision & Text Data • Model Evaluation & Reliability • Robustness & Generalization

Education

Doctor of Philosophy, Artificial Intelligence Oregon State University, Corvallis, OR, USA	<i>2021–Present</i>
Master of Science, Computer Science Iowa State University, Ames, IA, USA	<i>2016–2018</i>
Bachelor of Engineering, Computer Engineering Vidyalankar Institute of Technology, Mumbai, Maharashtra, India	<i>2014–2016</i>

Work Experience

Graduate Assistant <i>Location: Oregon State University</i> - TA for AI 534 (<i>Machine Learning</i>), CS 434 (<i>Machine Learning & Data Mining</i>), and CS 325 (<i>Analysis of Algorithms</i>) - Mentored students in algorithm design, data structures, supervised learning, model evaluation, and debugging best practices	Machine Learning <i>Sept 2021–Present</i>
Full Stack Software / ML Engineer <i>Location: Kingland, Iowa</i> - Co-created Kingland's enterprise Text Analytics platform, integrating deep-learning (NER), contextual language modeling, and document-parsing pipelines - Engineered cloud-native ingestion and NLP services that processed <i>5M+ financial records and 1.1M+ daily updates</i>, enabling large-scale extraction and normalization of entities from highly variable unstructured content - Achieved 99% entity-resolution accuracy for DTCC's mutual-fund data by designing context-aware extraction workflows addressing challenges such as alias resolution, domain-specific language, and bias mitigation - Built end-to-end features across the full stack — microservices, APIs, data pipelines, and UI components — ensuring seamless integration of ML models into production systems serving Fortune 500 clients	Text Analytics <i>July 2018–May 2021</i>
Summer Software Intern <i>Location: Icon Laboratory, Iowa</i> - Designed and implemented a security module for IoT devices, incorporating obfuscation and secure key-retrieval mechanisms to improve data privacy and system reliability - Integrated lightweight cryptographic techniques tailored for resource-constrained embedded IoT environments	IoT Security <i>May 2017 – Jul 2017</i>

Relevant Research Projects

MOSE-GNN: A Motif-Based Self-Explaining Graph Neural Network for Molecular Property Prediction <i>Oregon State University</i> - Designed MOSE-GNN, a graph neural network for molecular interaction modeling with built-in global explanations - Developed a molecular fragment-based interpretability framework producing <i>chemically meaningful</i> and <i>global</i> explanations - Demonstrated superior fidelity, consistency, and motif-correctness over post-hoc explainers - Enabled chemist-aligned model transparency for safety-critical molecular property prediction tasks	LoG conference <i>2024</i>
Graph Neural Networks for Learning Dynamics and Policy Generation with Explanations <i>Oregon State University</i> - Developed a graph neural network framework to learn system dynamics and generate control policies from structured state representations - Integrated decision trees as global explainers to extract human-interpretable rules governing learned policies - Demonstrated accurate forecasting of state transitions while preserving model transparency and causal structure - Implemented end-to-end pipelines in PyTorch, enabling reproducible experimentation on simulated dynamical systems	Research Project <i>2022</i>

Interpretable Deep Learning for Guided Microstructure–Property Exploration in Photovoltaics

Nature Partner Journals

Location: Iowa State University

2019

- Applied interpretable deep learning to identify **microstructural attributes** most predictive of photovoltaic efficiency.
- Built *explainable CNN pipelines* enabling scientists to trust, validate, and utilize ML-driven insights in materials discovery.
- Integrated attribution analysis with physics-based reasoning to accelerate photovoltaic materials exploration.

A Forward–Backward Approach for Visualizing Information Flow in Deep Networks

NeurIPS

Location: Iowa State University

2017

- Co-developed a **layer-wise attribution** method using mathematical inversion of the forward pass to analyze information flow
- Improved interpretability and debugging of CNNs, yielding a documented *15% improvement* in classification accuracy
- Demonstrated how forward–backward signal decomposition enhances transparency and reliability in deep neural networks

Additional Research (Selected)

Robustness studies, Reproducibility, and Interpretability of Deep Models

ISU repository & github

Multiple venues

2023–2025

- Studied adversarial robustness and failure modes of CNNs and Vision Transformers in medical imaging
- Analyzed training variability and reproducibility in deep neural networks; proposed reliability-aware evaluation metrics
- Conducted systematic evaluations of interpretability methods for convolutional image models
- Authored a position paper on domain-aligned evaluation standards for explainable AI

Leadership & Service

- **Ethics Reviewer** for the Dataset and Benchmarking Track, NeurIPS 2024 (USA)
- **Poster Presenter and Participant** in the MoML Conference & Hackathon, Valence Labs & Mila, 2024 (Canada)
- **Poster Presenter** at the OSU Research Showcase, Oregon State University, 2024 (USA)
- **Founder & Organizer** for Explainable AI reading group at Oregon State University, 2024 (USA)
- **Member & Speaker** at the Pervasive Personalized Intelligence Center (PPIC), 2023 (USA)
- **Participant** in the AgAID Hackathon, 2022 (USA)
- **Mentor** to three junior engineers, providing structured guidance in UI development and software testing, 2019 (USA)
- **Integration Testing Lead** across multiple Kingland projects, improving code coverage by *80%* and coordinating framework adoption across engineering teams in the US and China, 2020 (USA)
- **Honourable Mention** at the Kingland Poster Presentation, Iowa State University, 2018 (USA)
- **Chairperson** for student chapter of Computer Society of India (India)

Skills

Programming: Python, SQL, Java, C/C++, C#

Machine Learning: Statistical Modeling, Experiment Design, Model Evaluation, Explainable AI

Frameworks & Tools: PyTorch, TensorFlow, Keras, Pandas, NumPy, Scikit-learn

Systems: AWS, Docker, Terraform, Jenkins, Git, CI/CD Pipelines

Data: Large-scale data processing, feature engineering, analytics pipelines

Scientific ML: Graph Neural Networks, Time-Series Forecasting, Dynamical Systems, Simulation Modeling